

DNA Glycosylase Exercise

Background

In this exercise, you will explore the structure of a DNA repair protein found in most species, including bacteria. DNA repair proteins move along DNA strands, checking for mistakes or damage. DNA glycosylases, a specific type of DNA repair protein, recognize DNA bases that have been chemically altered and remove them, leaving a site in the DNA without a base. Other proteins then come along to fill in the missing DNA base.

Learning objectives

We will explore the relationship between a protein's structure and its function in a human DNA glycosylase called human 8-oxoguanine glycosylase (hOGG1).

Getting started

We will begin this exercise by exploring the structure of hOGG1 using a molecular 3-D viewer called StarBiochem. In this particular structure, the repair protein is bound to a segment of DNA that has been damaged. We will first focus on the structure hOGG1 and then on how this protein interacts with DNA to repair a damaged DNA base.

- To begin using StarBiochem, please navigate to: <http://mit.edu/star/biochem/>.
- Click on the **Start** button for **StarBiochem**.
- Click **Trust** when a prompt appears asking if you trust the certificate.
- Under **Samples**, select "DNA glycosylase hOGG1 w/ DNA – H. sapiens (1EBM)".

Take a moment to look at the structure from various angles by rotating and zooming on the structure.

- Instructions for changing the view of a structure can be found in the top menu, under **Help** → **Structure Viewing Instructions**.

The current way you are viewing the structure is by seeing each atom and bond in the protein drawn as a ball and a line, respectively. This way of representing a structure is called the **ball-and-stick** model and is the default model in StarBiochem. The **ball-and-stick** model allows you to see how atoms in the structure bond together. However, the space each atom occupies IS NOT accurately represented. To see a more realistic representation of the atoms in the structure you can use the **space-filled** model, where each atom is drawn as a sphere, whose size represents the physical space an atom occupies.

You can switch from the **ball-and-stick** model to the **space-filled** model in StarBiochem by increasing the size of the atoms in the structure:

- Notice that different atoms are slightly different in size.
- Gray = Carbon, Blue = Nitrogen, Red = Oxygen, Yellow = Sulfur
- Click on the **Primary** tab. The default atom size is 20% (**ball-and-stick** model).
- Move the **Atoms Size** slider to 100% (**space-filled** model).

The last page, Reference page, contains a series of terms and useful information that you will refer to during this exercise.

Exercise

hOGG1's Protein Structure Basics

1 Can you identify the DNA and the hOGG1 protein in the structure? Where is the DNA segment? Describe in your own words.

Answer

2 hOGG1 contains multiple sulfur atoms.

a) *Identify the name and sequence number of one of the amino acids in the structure that contains a sulfur atom.*

Answer

b) *Is the sulfur atom located in the backbone or in the side chain of the amino acid?*

Answer

3 Next, we will explore the primary structure (Reference page) of the hOGG1 protein. The hOGG1 protein consists of 325 amino acids. *List the 13 amino acids numbered 105 through 117 in order.*

Answer

105 _____	106 _____	107 _____	108 _____
109 _____	110 _____	111 _____	112 _____
113 _____	114 _____	115 _____	116 _____
117 _____			

4 Within a protein chain, amino acids form local structures called secondary structures (Reference page).

a) Explore the secondary structures found in hOGG1. *Are helices, sheets or coils present in hOGG1? Describe the color that represents each secondary structure you observe.*

Answer

Helices _____ (Yes or No) Sheets _____ (Yes or No) Coils _____ (Yes or No)

b) Amino acids 105 through 117 fold into one of the secondary structures. *Which secondary structure do they fold into?*

Answer

5 Now we will explore the relationship between DNA glycosylase's structure and one of the several types of amino acids that contribute to hOGG1's overall shape, its tertiary structure (Reference page).

a) Negatively charged amino acids are hydrophilic (Reference page). ***Are the negatively charged amino acids located on the inside (buried) or outside (exposed) of this protein? What does that suggest about the cellular environment surrounding this protein, is it hydrophobic or hydrophilic?*** Explain your answer.

Answer

hOGG1's Interaction with DNA

6 Now let's explore how DNA glycosylase interacts with DNA to recognize the damaged DNA base within its sequence. DNA is composed of four bases: Adenine (A), Thymine (T), Cytosine (C) and Guanine (G). In this particular structure, the hOGG1 protein is bound to a segment of DNA that contains an oxidized guanine.

Take a closer look at how the DNA bases are oriented within the double helix. Each base is attached to a sugar and phosphate backbone to form a complete nucleotide. Within the double helix, a base within one strand pairs with another base from the opposite strand by hydrogen bonding forming a base pair: Adenine (A) base pairs with Thymine (T) and Cytosine (C) base pairs with Guanine (G).

a) How many DNA base pairs can you count within this double helix?

Answer

b) How many DNA bases are unpaired (not paired to its partner on the other strand)?

Answer

c) Is the oxidized guanine ([8OG]25) base paired or unpaired? Describe the position of the oxidized guanine with respect to the hOGG1 protein and the double helix. What does this suggest about the mechanism that hOGG1 uses to identify damaged DNA bases?

Answer

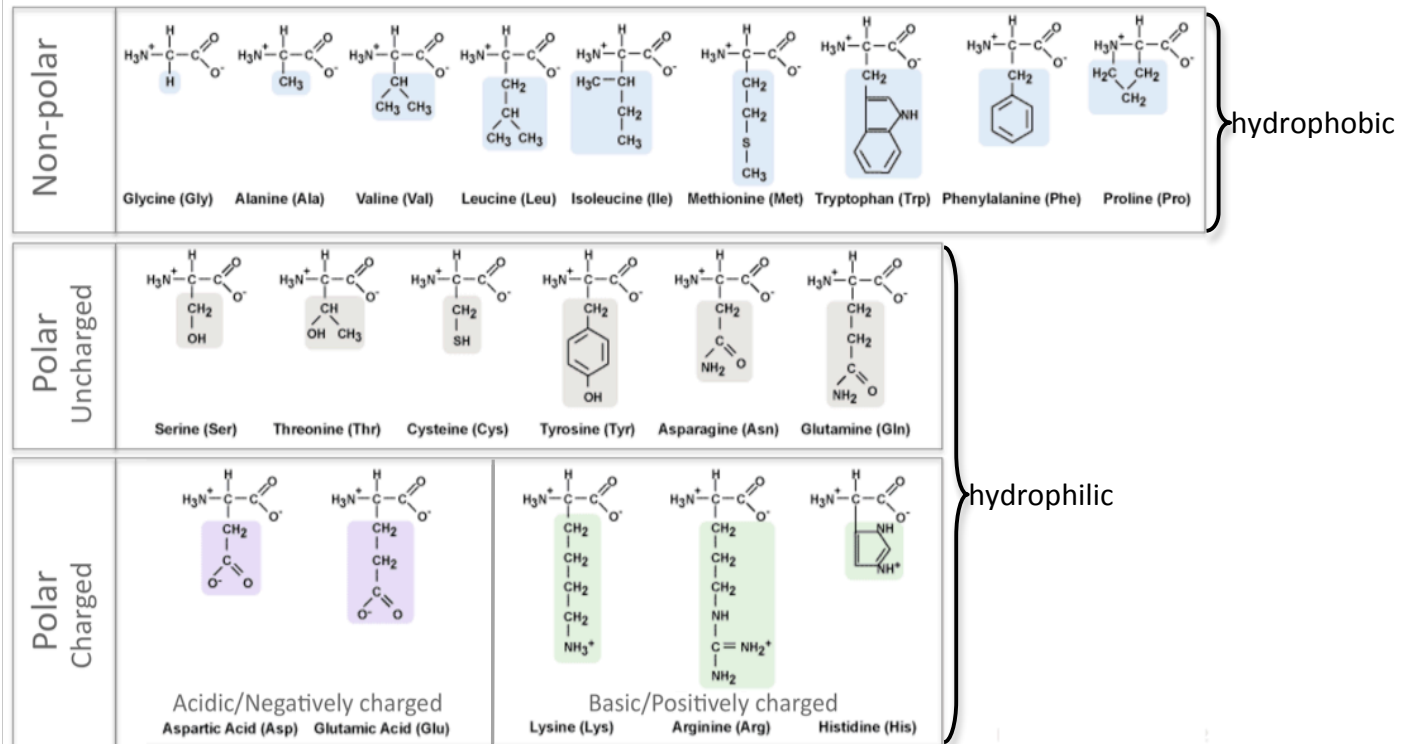
7 Certain amino acids within hOGG1 form contacts with the DNA and are able to recognize if a guanine base has been damaged by oxidation. *Where are you more likely to find the amino acids that recognize damaged guanine bases within hOGG1, in Helix 1 or Helix 16?* Explain why.

Answer

Reference

CHEMICAL STRUCTURES OF THE AMINO ACIDS

The 20 amino acids share a common backbone and are distinguished by different side chains, also called 'R' groups, highlighted by the various colors below.



PROTEIN STRUCTURE BASICS

All proteins have the following three levels of protein structure:

Primary structure

Describes the order of the amino acids in the protein chain but does not describe its shape.

Secondary structure

Describes shapes that form from local folding of regions within the amino acid chain. These smaller structures can be divided into two main types: helices and sheets. Coils are made of amino acids that do not form regular secondary structures (helices and sheets) but play important roles in protein folding.

Tertiary structure

Describes the entire folded shape of a protein chain.

In addition, some proteins interact with themselves or with other proteins to form larger protein structures. These proteins have an additional level of protein structure:

Quaternary structure

Describes how multiple protein chains interact and fold to form a larger protein complex.